White Flight and Residential Segregation in Mobile County, AL: Aligning Historical Indicators and U.S. Census Bureau Data to Model Dissimilaritiy

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Abstract

This study analyzes demographic shifts in Mobile County, AL from 1960 to the present to assess the persistence of residential segregation. By utilizing decennial census data and spatial analysis, the research examines how structural and historical forces continue to drive spatial inequality in the post–civil rights era.

Overview

White Flight and Segregation. White flight occurs when white individuals abandon integrated spaces to avoid growing minority populations and stall racial progress. This migration is driven by the desire to protect "whiteness as property," where distancing themselves from other races preserves their social and economic capital (Crowder & South, 2008). In Mobile, desegregation efforts were only made when they protected the elite's economic interests and established new, spatial forms of white exclusivity (Bell, 1980).

Segregation Facilitators.

- *Whiteness as property*: Whiteness is not just viewed as a racial identity, but as a legal and social asset that operates at a systems level. It posits that European ancestry carries a distinct set of inherent privileges, sociopolitical status, and expectations of entitlement that function as a protected form of property (Donner, 2021).
- *Interest Convergence*: Racial equality is achieved when the interests of Black people align with the self-interests of a white majority (Bell, 1980).
- *Antiblackness*: We also bridge in contemporary discussions from the field of Black studies regadring conceptions of antiblackness.

Literature Review

- Between 1970-1980, Mobile County saw a significant decline in public school enrollment (approx. 11,500 students), directly correlating with a rise in private school attendance and suburban migration (Pride, 2022). This period highlights a distinct pattern of "white flight."
- Research indicates that white residents often undervalue racially integrated areas due to an "aversion to living near black neighbors" and inflated perceptions of crime (Crowder & South, 2008). These subjective assessments frequently override objective neighborhood quality, driving residential turnover.

Research Questions

How have Black and white racial demographics across tracts in Mobile County evolved from 1980-2020?

Mathematical Model

$$D = \frac{1}{2}\sum_{i=1}^{n}\left|\frac{b_i}{B} - \frac{w_i}{W}\right|$$

where

- N is the number of census tracts,
- b_i is the population of Black residents in unit i,
- B is the total population of Black residents in the geographic area,
- w_i is the population of white residents in unit i,
- W is the total population of white residents in the geographic area.

Hypothesis

Levels of residential segregation will remain persistently high, reflecting the continued influence of structural factors such as white flight and path-dependent patterns of residential sorting.

Limitations

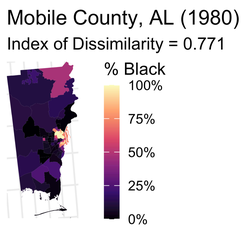
A significant limitation of this analysis was the inability to analyze NHGIS data from 1960, and 1970. Accessing these decades would have provided critical insights into the baseline of segregation before and during the most volatile periods of the Civil Rights Movement and initial court-ordered desegregation.

Methods and Data

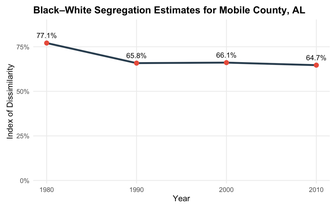
This study uses historical census data from the U.S. Census Bureau to examine changes in Black and white residential distribution across census tracts in Mobile County from 1960 to 2010. Data are analyzed using spatiotemporal demographic mapping, involving longitudinal comparison of tract-level racial composition and the calculation of segregation indices across decennial census years. Our interpretation of observed trends is informed by a Critical Race Theory framework, particularly the concepts of interest convergence and whiteness as property, to contextualize segregation patterns over time.

Findings

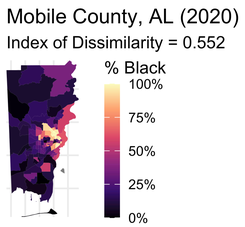
For 1980, $D = 0.771$.



We use the spatiotemporal analysis to track changes in D from 1980 to 2020. We find that segregation remained high with some slight, but significant, decreases due to public policies.



We separate data for 2020 due to Covid-19.



Discussion

The 11.3% decrease in the Dissimilarity Index between 1980 and 1990 possibly represents a period of transitory diversity, rather than sustainable integration. As white families abandoned the urban core, census tracts briefly appeared integrated before reverting to new patterns of segregation.

Our data reveals a disconnect between legal milestones and spatial reality. While MCPSS achieved "unitary" status in 1997 , the 2020 Index of Dissimilarity remains high at 0.552. This suggests that while legal oversight ended, private-sector white flight and path-dependent sorting persisted as mechanisms to maintain racial distance and preserve whiteness as property.

The 1988 Mobile Plan exemplifies Interest Convergence. By establishing elite magnet schools like Council Traditional and Phillips Preparatory, the county incentivized white families—who had withdrawn 11,500 students the prior decade—to remain in the system. This protected "whiteness as property" by offering exclusive educational assets in exchange for technical legal compliance, effectively stalling deeper residential integration.

The 2020 estimate reflects a "frozen" snapshot. The pandemic stalled residential mobility and disrupted Census operations in high-density tracts. Consequently, the 0.552 index may mask the true contemporary movement of residential sorting, as lockdowns temporarily suppressed the migration patterns our model tracks.

Conclusion

Our analysis reveals that residential segregation in Mobile County is a persistent structural force. To deepen this understanding, future research should extend the longitudinal dataset to 1950 and introduce qualitative interviews to capture the lived experiences and oral histories that quantitative data alone cannot reveal.

Acknowledgements

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References

Scan to see our timeline and references!